

KARTIK GAHLOT

[GMAIL](#) | [CONTACT NO.](#) | [LINKEDIN](#) | [LEETCODE](#) | [GITHUB](#)

EDUCATION

B.Tech	2024-2028	NSUT	8.57 CGPA
CBSE (Class XII)	2023	ITBP Public School	92.4%
CBSE (Class X)	2021	ITBP Public School	95.6%

TECHNICAL SKILLS

Languages : Python, C++, HTML, CSS, JavaScript, React, Node.js, SQL, Assembly
Domains : Machine learning, Deep learning, Computer Vision, Web Development
Tools & Databases : Git, Github, Linux, VS Code, MySQL, MongoDB
Core CS : Data Structures & Algorithms, Object Oriented Programming, DBMS, System Design

PROJECTS

CONTENT BASED BOOK RECOMMENDATION SYSTEM ([Github Repository](#))

- Built a machine learning recommendation system using content based filtering achieving 94.3% accuracy, trained on book metadata (genres, authors, publishers) using cosine similarity to suggest relevant books, served through a FastAPI backend with a real time search frontend.
- Designed a custom evaluation framework to measure model quality without labeled data, computed metrics including Catalog Converage, Diversity, Personalization, and Novelty to validate recommendation accuracy
- Tech used : Python, Machine learning, HTML, CSS, Vanilla JavaScript, FastAPI

EMOTION RECOGNITION USING ELECTROENCEPHALOGRAM (EEG) ([Github Repository](#))

- Built deep learning models for brain computer interface tasks using TensorFlow and Scikitlearn, implemented emotion recognition on the DEAP EEG dataset and brain tumor classification on MRI scans using CNN and a hybrid CNN & SVM architecture
- Developed a segmentation model on the BraTS20 dataset to detect and localize brain tumors from MRI scans applying the same spatial feature extraction principles used in medical imaging and perception systems.
- Tech used : Python, Scikit learn, TensorFlow/Keras, Hybrid CNN & SVM, Numpy, Pandas, Matplotlib

PLANT DISEASE DETECTION SYSTEM ([Github Repository](#))

- Engineered a CNN based image classification model using TensorFlow/Keras trained on the PlantVillage dataset available on Kaggle across 38 plant disease classes and 54,000+ leaf images, implemented a deep architecture with Conv2D, Batch Normalization, MaxPooling, and Dropout layers with Softmax output achieving robust multi class disease detection from leaf images.
- Built a custom label parser to map raw model predictions to human readable plant disease outputs, visualized model performance via confusion matrix, ROC curves, precision recall curves, and per class accuracy graphs.
- Tech used : Deep Learning, CNN, Python, Keras, Matplotlib, Seaborn, Numpy, Pandas

LEADERSHIP EXPERIENCE

As **Tech Lead** of The Debugging Society - Biggest Tech Society of NSUT

- Led a 180 member tech society, organised 4-5 events & hackathons and onboarded 80 recruits

As **Sponsorship Head** in Innovision '25 - Tech fest of NSUT

- Secured 5-6 brand sponsorships for NSUT's largest tech fest with 2000+ attendees

As **Volunteer** in NSS (National Seva Sangh) - Social Welfare Society

- Organized social service initiatives, awareness drives, and community welfare activities as an NSS Volunteer.

ACHIEVEMENTS AND CERTIFICATIONS

- Won GenIgnite Hackathon held at Microsoft Office Noida and won a few other University level Hackathons
- Microsoft Azure AI essentials certification ([Link](#))
- Hackerrank Software Engineer Intern certification ([Link](#))
- Qualified for Round 2 of Smart India Hackathon 2025
- Secured 99.43%ile in JEE Mains 2024 and qualified JEE Advanced 2024